Varun Sureka

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Education

Carnegie Mellon University **Pittsburgh, PA**
Master of Science in Mechanical Engineering Aug. 2026 – May 2027
Bachelor of Science in Mechanical Engineering GPA: 3.8/4.0 May 2026
Relevant Coursework: Medical Devices, Bioinspired Robot Design, Applied Finite Element Analysis, Modern Control Theory, Heat Transfer, Dynamic Systems, Fluid Mechanics, Robot Dynamics and Analysis

Experience

Mechanical Engineering Intern - Strategic Robotics R&D **May 2026 – August 2026**
Johnson & Johnson MedTech | *Solidworks* Santa Clara, CA

- Prototyped a robotic pump capable of precisely dosing 8 fluids whose viscosities spanned 0–150 cP, including shear-thinning fluids, packaged within a 1x1x1" form factor.
- Validated pump headloss and flow rate via hand calculations, comparing predictions against benchtop testing and off-the-shelf alternatives to inform pump selection.
- Designed custom test fixtures to benchmark off-the-shelf instruments. Authored test protocols and reports informing development of DIRs and SSSs for next-gen instrumentation.

Mechanical Engineering Co-Op **May 2025 – December 2025**
CTL Amedica | *PTC Creo, Fusion 360* Philadelphia, PA

- Redesigned a line of paddle distractors per surgeon feedback, adding cannulation and reducing overall mass by 10%.
- Performed mechanical validation and verification to ensure instruments could withstand torsional and shear loading conditions. Designed test fixtures and wrote engineering test protocols and reports documenting the validation process.
- Supported sustaining engineering efforts by updating CAD models, drawings, and component designs for existing instrumentation following standard design change and document control processes.

Projects

Automated Spot Welder Gantry System | *SolidWorks, MATLAB* **Spring 2026**

- Awarded best overall project for engineering a 3-axis CNC gantry capable of executing complex weld sequences within a ±0.15mm XY positional tolerance.
- Developed MATLAB torque-speed simulations to size stepper motors, validating predicted time-to-position within 0.01s across 15+ validation trials.
- Designed a controller-driven interface supporting manual jog control, live waypoint recording, and automated multi-point path replay for repeatable hands-free welding operations.

Inverter Repackage System Lead - Carnegie Mellon Racing (FSAE) | *Solidworks* **May 2023 – May 2024**

- Led mechanical design and simulation of a liquid coldplate for EV inverter cooling, reducing surface temperature differential by 5°C.
- Achieved a 50% mass reduction (1500g), improving heat transfer and mechanical performance.
- Performed thermal and structural validation using SimScale and Ansys across varied loads and flow rates.

Whiteboard Drawing with Franka Arm | *Python, Git, FrankaAPI* **Fall 2024**

- Awarded best project for implementing a motion control system on a 7-DOF Franka Emika arm to draw detailed whiteboard images with continuous-line planning.
- Developed velocity profiles and motion constraints for smooth, collision-free trajectories.

SLAM Project: Particle Filtering Using Monte Carlo Localization | *Python, Git* **Spring 2025**

- Built a particle filter with adaptive resampling to localize a mobile robot within a known occupancy grid, using Monte Carlo methods to estimate pose under noisy motion and sensor data.
- Modeled probabilistic odometry and range sensor measurements, dynamically adjusting the number of particles based on uncertainty, and visualized the real-time belief evolution using Matplotlib.

Leadership / Extracurricular

Teaching Assistant - Intro to Robotics **Spring 2024, Spring 2025, Spring 2026**
Carnegie Mellon University

- Instructs students on fundamental robotics concepts including control systems, planning, localization, and robotic design.

Technical Skills

Mechanical Design & Analysis: Soldiworks, PTC Creo, Onshape, Fusion 360, Ansys Discovery, SimScale
Programming & Simulation: Python, MATLAB, C, C++, ROS2